RAG Learning

P5 — Energy

OCR GCSE Physics A (J249) — Gateway Science · Paper 2

P5.1 Energy Stores and Transfers

Energy is never created or destroyed — it is only **transferred** from one store to another (or from one object to another). This is the **principle of conservation of energy**.

The Eight Energy Stores

Store	Where it is	Example
Kinetic	In anything moving	A moving car, a falling apple
Gravitational potential	In objects at height	A raised weight, water in a dam
Elastic potential	In stretched / compressed springs	A drawn bow, a compressed spring in a toy car
Thermal (internal)	In warm objects — random motion of particles	A hot cup of tea
Chemical	In bonds between atoms	Fuel, food, a battery
Magnetic	In separated magnetic poles	Two magnets held apart
Electrostatic	In separated electric charges	A charged balloon near a wall
Nuclear	In the nuclei of atoms	Uranium fuel in a reactor

© RAG Learning · Not for redistribution

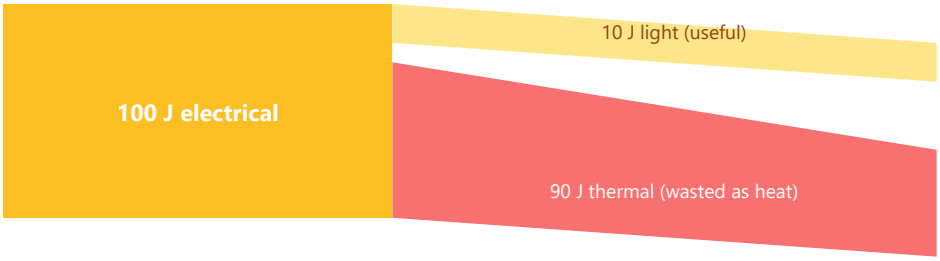
Energy Transfer Pathways

Energy is moved between stores by four **pathways**:

1. **Mechanically** — a force acts through a distance (work done).
2. **Electrically** — charge moves through a pd.
3. **By heating** — temperature difference drives energy flow.
4. **By radiation** — waves (light, sound, IR) carry energy.

Worked Example: A kettle: chemical store (gas / electricity source) → electrical pathway → thermal store of water.
A falling ball: gravitational store → kinetic store (as it falls) → thermal store (of ground + ball when it hits).
A torch: chemical store (battery) → electrical pathway → thermal + light radiation stores (bulb).

Sankey Diagram for a Filament Lamp (100 J in)



Widths proportional to energy. Notice most energy is wasted in an old filament bulb.

P5.2 Calculating Energy

Kinetic Energy

$$E_k = \frac{1}{2} \times m \times v^2 \quad (J) = (kg) \times (m/s)^2$$

Note: doubling speed *quadruples* kinetic energy (because of v^2).

Worked Example: A 1 200 kg car travels at 20 m/s. Kinetic energy?
$E_k = ½ \times 1200 \times 20^2 = ½ \times 1200 \times 400 =$ **240 000 J = 240 kJ**.
At 40 m/s (double): $E_k = ½ \times 1200 \times 40^2 = 960$ kJ — four times as much.

Gravitational Potential Energy

$$\mathbf{E_p = m \times g \times h} \quad (J) = (kg) \times (N/kg) \times (m)$$

$g \approx 9.8$ N/kg on Earth (often rounded to 10).

Worked Example: A 70 kg climber ascends a cliff 150 m high. GPE gained?
$E_p = m \times g \times h = 70 \times 9.8 \times 150 =$ **102 900 J ≈ 103 kJ**. This is the minimum work the climber must do against gravity.

Energy Stored in a Spring (Elastic Potential)

$$\mathbf{E_e = ½ \times k \times e^2} \quad (J) = (N/m) \times (m)^2$$

k = spring constant, e = extension.

Energy for Heating — Specific Heat Capacity

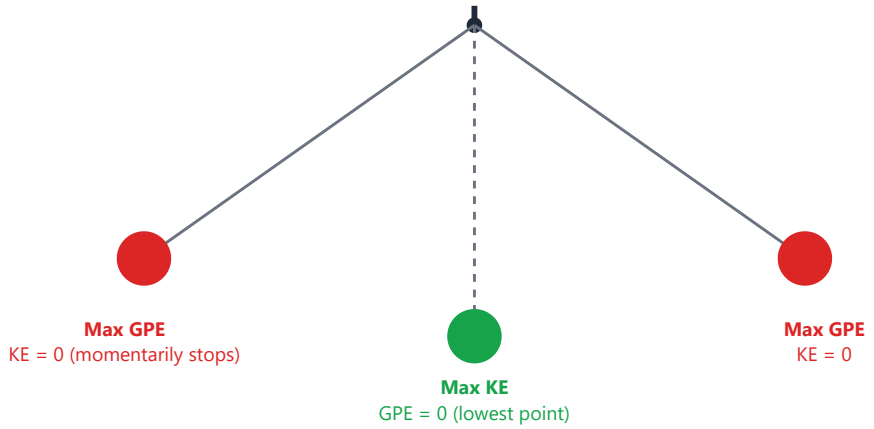
$$\mathbf{\Delta E = m \times c \times \Delta\theta} \quad (J) = (kg) \times (J/kg\ °C) \times (°C)$$

Covered in detail in P1; worth re-reading as it often appears in energy-efficiency questions.

P5.3 Energy Transfers in a Closed System

In a closed, **frictionless** system, as one energy store goes down another goes up by exactly the same amount (conservation of energy). The total is constant.

Pendulum — GPE ↔ KE Conversions



Worked Example: A 0.5 kg pendulum bob is released from a height of 0.8 m above its lowest point. Speed at the bottom (no friction)?
Energy conservation: all GPE → KE.
$m\,g\,h = \frac{1}{2}\,m\,v^2 \;\rightarrow\; v = \sqrt{(2\,g\,h)} = \sqrt{(2 \times 9.8 \times 0.8)} = \sqrt{15.68} =$ **3.96 m/s**.

P5.4 Efficiency and Power

Efficiency is how good a device is at transferring energy to useful output. It is a ratio (0 to 1) or a percentage.

$$\text{Efficiency} = \text{useful output} / \text{total input} \times 100\%$$

$$\eta = E_{useful} / E_{total} \times 100\% \quad \text{or} \quad \eta = P_{useful} / P_{total} \times 100\%$$

Worked Example: A motor uses 500 J of electrical energy and lifts an object giving it 320 J of GPE. Efficiency?
$\eta = 320 / 500 \times 100\% =$ **64%**. The rest (180 J) is wasted — mostly as heat in the motor.

Power

Power is the rate of energy transfer — how much energy per second. 1 watt = 1 joule per second.

$$P = E / t \quad (W) = (J) / (s) \quad \text{or} \quad P = W / t \text{ (power = work done per unit time)}$$

Worked Example: A crane lifts a 2 000 kg load through 15 m in 25 s. Output power?
Work done = m g h = 2000 × 9.8 × 15 = 294 000 J.
P = 294 000 / 25 = **11 760 W ≈ 11.8 kW**.

P5.5 Reducing Unwanted Energy Transfers

In every real process some energy is transferred to **less useful stores** (usually thermal), often because of **friction** or **air resistance**. Good engineering minimises the fraction lost.

Device / situation	Wasted energy goes to	Ways to reduce waste
Car engine	Thermal (friction in engine & tyres, air resistance, exhaust heat)	Lubrication, low-friction bearings, aerodynamic shape, regenerative braking
Old filament bulb	Thermal (heat from hot filament)	Replace with LED — far higher efficiency
Home heating	Thermal loss through walls, roof, windows	Loft insulation, cavity wall insulation, double glazing, draught proofing
Speaker	Thermal (coil heating)	Better magnet materials, efficient amplifier

Insulation & Thermal Conductivity

The rate of thermal conduction through a wall depends on the **thermal conductivity** of the material and its **thickness**. A low-conductivity material (good insulator) resists heat flow and keeps a house warm.

Exam Tip: Double glazing: two panes of glass with a small air or argon gap. Air has very low thermal conductivity, so heat conducts very slowly across the gap, reducing loss through windows by about half compared with single glazing.

P5.6 National & Global Energy Resources

A **renewable** energy resource is one that is replenished as fast as it is used — it will not run out on a human timescale.
A **non-renewable** resource is finite — burning it uses it up faster than natural processes can form more (e.g. fossil fuels took hundreds of millions of years to form).

Energy Resources — Renewable vs Non-renewable

RENEWABLE

- **Solar** — Sun's energy → electricity via PV cells / heat water
- **Wind** — Turbine spins generator; needs windy site
- **Hydroelectric** — Dammed water drives turbines; uplands
- **Tidal** — Tides move barrage turbines; predictable
- **Wave** — Waves compress air → drives turbine
- **Geothermal** — Earth's interior heat; limited geography
- **Biofuels**

NON-RENEWABLE

- **Oil** — Transport fuels, plastics; CO_2 emissions
- **Coal** — Electricity generation; high CO_2 & SO_2
- **Natural gas** — Heating, electricity; cleanest fossil fuel
- **Nuclear (uranium)** — Low CO_2; radioactive waste challenge

Benefits vs costs
Fossil fuels: reliable + cheap; burning releases CO_2 (global warming) & pollutants.
Nuclear: high output, but waste storage is a long-term problem; accident risk.

Comparing Energy Resources

Resource	Reliability	CO_2	Environmental impact
Coal	Very reliable	High	Mining scars, smog, acid rain (SO_2)

Natural gas	Very reliable	Medium	Less than coal; methane leaks serious
Nuclear	Very reliable (base-load)	Very low	Radioactive waste, accident risk
Solar	Not at night; depends on weather	Very low	Land use; some rare materials in panels
Wind	Only when it is windy	Very low	Visual impact, noise, bird strikes
Hydroelectric	High if reservoir full	Very low	Floods valleys, disrupts ecosystems
Tidal	Very predictable	Very low	Disturbs estuarine wildlife
Wave	Variable	Very low	Affects shipping, coastal wildlife
Biofuel	Reliable	Near-zero net (if replanted)	Competes with food production
Geothermal	Very reliable	Very low	Geographic limits (Iceland, Kenya)

The UK Energy Mix & Why It's Changing

The UK has been **moving away from coal** (virtually zero in 2025 compared with over 40% in the 1990s). Reasons:

- **Climate change** — commitments under the Paris Agreement and UK Net Zero target for 2050 require cutting CO_2 emissions.
- **Air quality** — coal emits SO_2, NO_X, particulates, all harmful to health.
- **Falling renewable costs** — wind and solar are now often cheaper per kWh than new-build gas.
- **Energy security** — domestic renewables reduce reliance on imported fuel.

Balancing Supply and Demand

Demand varies daily (peaks in the evening) and seasonally (higher in winter). Grid operators need a mix of:

- **Base-load** — runs 24/7 (nuclear, biomass, some gas).
- **Dispatchable** — can be turned up or down quickly (gas, hydro).
- **Variable renewables** — when available (wind, solar).
- **Storage** — pumped-storage hydroelectric and battery farms absorb surplus and release it when demand peaks.

> **Worked Example: Dinorwig pumped-storage station (Wales)** — off-peak surplus electricity pumps water uphill into a reservoir (GPE stored). At times of peak demand, water flows back down through turbines (GPE → electrical) in just seconds. Useful as a grid "battery" with rapid response.

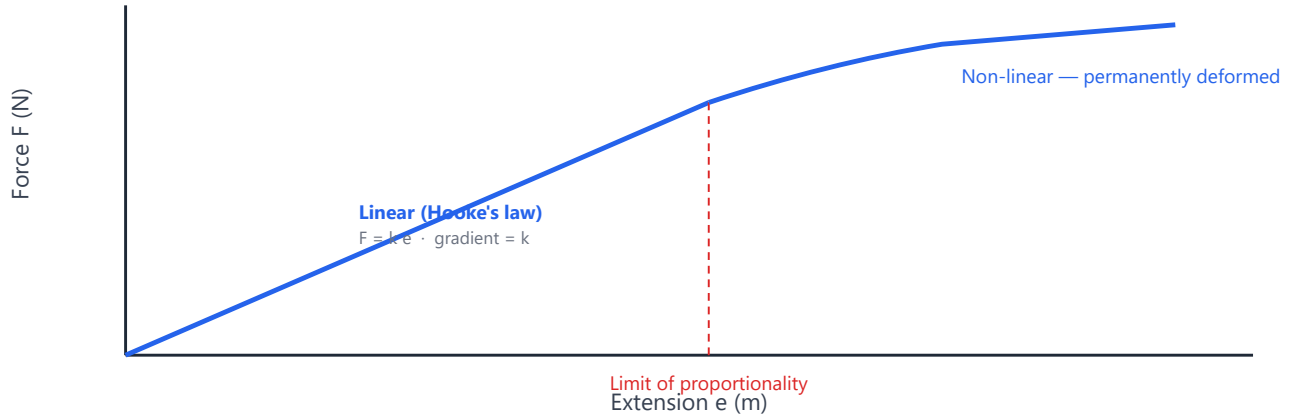
P5.7 Forces and Springs (Work Done on Springs)

When a force stretches or compresses a spring, work is done, and energy is stored in the **elastic potential** store.

$$\mathbf{F = k \times e} \text{ (Hooke's law)}$$

F = force (N), k = spring constant (N/m), e = extension (m). Valid below the *limit of proportionality*.

Force–Extension Graph for a Spring



Up to the **limit of proportionality**, extension is directly proportional to force (straight line through origin). Beyond this, the spring deforms permanently (**plastic deformation**).

Worked Example: A spring extends 6.0 cm (0.06 m) when a 3.0 N weight is hung on it. Spring constant?
$k = F / e = 3.0 / 0.06 =$ **50 N/m**.
Elastic PE stored: $E_e = ½ k e^2 = ½ \times 50 \times 0.06^2 =$ **0.09 J**.

Quick Summary — Key Equations for P5

Quantity	Equation	Units
Kinetic energy	$E_k = ½ m v^2$	J
Gravitational PE	$E_p = m g h$	J
Elastic PE	$E_e = ½ k e^2$	J
Heating	$\Delta E = m c \Delta\theta$	J
Work done	$W = F d$	J
Power	$P = E / t$	W
Efficiency	$\eta = E_{useful} / E_{total} \times 100\%$	%
Hooke's law	$F = k e$	N

© RAG Learning · Not for redistribution